

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Industry Problem Solving Task

Criteria	Understandin g of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solutio n Design (2.5)	Use of Tools (2)	Analysis (1)	Total(10)
<i>Weightag e</i>	20%	25%	25%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

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Annexure A

Q1 . Dataset: Supermarket Purchases

Customer Visits/Month Avg Spend (₹) Items Bought

C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. *(BL3 – 7 Marks)*

Q2. Dataset: Credit Card Transactions

Transaction Amount (₹) Location Time (hrs)

T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. *(BL4 – 7 Marks)*

Q3. Dataset: Telecom Usage

User Call Duration (min) Data Usage (GB) Recharge (₹)

U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question:

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. *(BL4 – 7 Marks)*

Q4 . Dataset: E-commerce Behavior

User Product Views Clicks Purchases

U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:

Suggest a clustering technique to group users and explain how it improves recommendation systems. *(BL3 – 7 Marks)*

Q5. Dataset: Traffic Data

Location Vehicle Count Avg Speed (km/h) Delay (min)

L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. *(BL4 – 7 Marks)*

Q1. ANSWER:

Use K-Means clustering with the features Visits/Month, Average Spend and Items Bought. Standardize the three features first because their scales differ. Choose $k = 2$ or 3 using the Elbow method. For this dataset, one cluster can represent frequent, lower/moderate-spend customers such as C1, C3 and C5, while another can represent less-frequent, higher-spend customers such as C2 and C4. The clusters can support targeted marketing: frequent customers can receive loyalty rewards and personalized offers, while high-spending customers can receive premium plans, exclusive products and retention offers. Cluster profiles should be validated using measures such as silhouette score.

Q2. ANSWER:

Use clustering on Transaction Amount, Location and Time after suitable encoding and scaling. K-Means or DBSCAN can be used; DBSCAN is useful for identifying sparse points as outliers. Normal transactions are expected to form dense groups, while unusual transactions such as very high amounts at unusual times or locations may appear as separate clusters/outliers. In the sample, T2 and T4 have much higher amounts and occur at 2 and 1 hours, so they should be investigated as potentially unusual transactions. The system can flag such records for additional verification. Clustering is a screening method and should be combined with transaction history and other fraud rules before declaring a transaction fraudulent.

Q3. ANSWER:

Apply K-Means clustering using Call Duration, Data Usage and Recharge after standardization. The clusters can represent high-usage/high-recharge customers, moderate users and low-usage/low-recharge customers. In this dataset, U2 and U4 form a low-usage group, while U1, U3 and U5 form a high-usage group. A churn-risk model can prioritize users whose usage or recharge is declining or who belong to a low-engagement cluster. Retention strategies can include personalized data plans, recharge benefits, discounts and proactive communication. Clustering helps identify customer segments, but churn prediction should also use historical churn labels when available.

Q4. ANSWER:

Use K-Means clustering with Product Views, Clicks and Purchases after scaling the features. Users with similar browsing and purchasing behaviour are grouped together. In the dataset, U1, U3 and U5 show relatively high activity and purchases, U2 shows moderate activity, and U4 shows low activity. For each cluster, the recommendation system can learn popular products and preferences and recommend relevant items to users in that cluster. This improves recommendation quality by using behavioural similarity rather than treating every user identically. Cluster performance can be evaluated using silhouette score and recommendation metrics such as precision or click-through rate.

Q5. ANSWER:

Use K-Means or DBSCAN clustering on Vehicle Count, Average Speed and Delay after scaling the features. Congested locations should show high vehicle count, low average speed and high delay. In the dataset, L2 and L4 clearly form a high-congestion group, while L1, L3 and L5 form a lower-congestion group. The congestion clusters can be displayed on a traffic map and used to prioritize signal optimization, alternate routes, road improvements and traffic-control measures. Real-time data can be clustered continuously to detect changing congestion patterns and support dynamic traffic management.